Ajay Payer

Mumbai, Maharashtra | +91 8669180821 | ajaypayer5@gmail.com | ajay-payer | ajaypayer

Education

Veermata Jijabai Technological Institute (VJTI) 2025 – Present
Bachelor of Technology in Computer Engineering

Dr. Babasaheb Ambedkar Technological University (DBATU) 2022 – 2025
Diploma in Information Technology | 99.30% | State Rank: 11

New English School, Kharsai 2022
Secondary School Certificate (SSC) | 88.00%

Experience

Konect U | *Generative AI / Machine Learning Intern* June 2024 – July 2024

- Developed a **machine learning solution** for **Air Dust Particle Size Measurement** using **Python** and environmental datasets.
- Performed **data preprocessing**, **exploratory data analysis (EDA)**, **feature engineering**, and **model evaluation** using **Pandas**, **NumPy**, and **Matplotlib**.
- Implemented and benchmarked **4 regression models**– **Multiple Linear Regression**, **Random Forest**, **SVR**, and **XGBoost**– to identify the **best-performing approach**.

Projects

HTE-Based AI Decision Platform | React, Django, SQLite, ChromaDB, BM25, Groq | Aug 2026

- Developed a centralized **decision-support platform** analyzing **11 educational datasets** across **20+ engineering colleges** for **institutional analytics**, **college comparison**, **enrollment forecasting**, and **executive reporting**.
- Built an **AI decision assistant** that answers natural-language queries using **structured institutional data** and **document-based RAG** for **source-grounded insights** and college comparisons.
- Implemented **branch-wise enrollment forecasting** using **engineered features**, benchmarking **ML algorithms** and providing **prediction confidence** and **feature-level explanations**.

Krushi Sarthi | Python, Django, JavaScript, SQLite, Gemini | Feb 2026 – May 2026

- Developed a centralized portal covering **3 key areas**– **government schemes**, **MSP**, and **agricultural loans**–with eligibility details and application information.
- Added an **AI chatbot** and **multilingual support** in **3 languages**–English, Hindi, and Marathi–to improve access to agricultural information.
- Implemented **farmer registration**, **admin-based content management**, and **automated SMS alerts** for newly published schemes and MSP updates.

Air Dust Particle Size Measurement | Python, Pandas, NumPy, Scikit-learn, Matplotlib | Jun 2024 – Jul 2024

- Developed a **machine learning solution** to predict **PM2.5 concentration** using **2 environmental datasets** from reference and low-cost sensor sources.
- Performed **data cleaning**, **timestamp processing**, **hourly averaging**, **dataset merging**, **feature engineering**, and **IQR-based outlier detection** to prepare the data for modeling.
- Compared **4 regression models** using **MAE**, **RMSE**, and $\mathbf{R^2}$; **Random Forest** achieved an $\mathbf{R^2}$ **of 89.2%**, with **RMSE 43.94** and **MAE 28.86**, making it the best-performing model.

Relevant Coursework

- Data Structures & Algorithms
- Object Oriented Programming
- Operating Systems
- Software Engineering
- Database Management Systems
- Automata Theory

Skills

Technical Skills: C, C++, Java, Python, JavaScript, HTML, CSS.
Databases: MySQL, SQLite.
Libraries & Frameworks: Django, Spring Boot (Basic), Pandas, NumPy, Scikit-learn, Matplotlib.
Tools: Git, GitHub, VS Code, Google Colab, Jupyter Notebook, Android Studio, IntelliJ IDEA, Eclipse, Postman.

Achievements

- Developed an **AI-powered HTE Decision Intelligence Platform** for adoption by the Government of Maharashtra to enable data-driven higher education decision-making
- Secured **99.30%** in Diploma in Information Technology and achieved **State Rank 11** in Maharashtra.
- Recognized as the Top Academic Performer of the Year in the Information Technology Department during the Diploma program.

Certifications

- Git/GitHub
- NVIDIA – Accelerating End-to-End Data Science Workflows
- Lenovo LEAP NextGen Scholar Program in Artificial Intelligence